ELC -2026 FINAL EVALUATION PRESENTATION

Ground Resolution Enhancement for Lower Earth Orbit Satellites

Team Members

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MOTIVATION

- LEO satellite images often have limited ground resolution. High-resolution sensors are expensive and hardware-limited.
- Software-based enhancement can improve image quality for Earth observation applications.

OBJECTIVE

- The objective of this project is to enhance the ground resolution of LEO satellite images using computer vision techniques like image stacking and super-resolution methods to reconstruct higher-quality images while preserving important spatial details.
- Evalute the effectiveness of these methods using quality metrics.

fig. Resolution Enhancement Through Stacking

Swath Analysis

- Studied swath geometry for estimating image coverage.
- Computed swath width using satellite imaging parameters.
- Evaluated the coverage–resolution (GSD) trade-off.
- Integrated swath calculations into the geolocation pipeline.

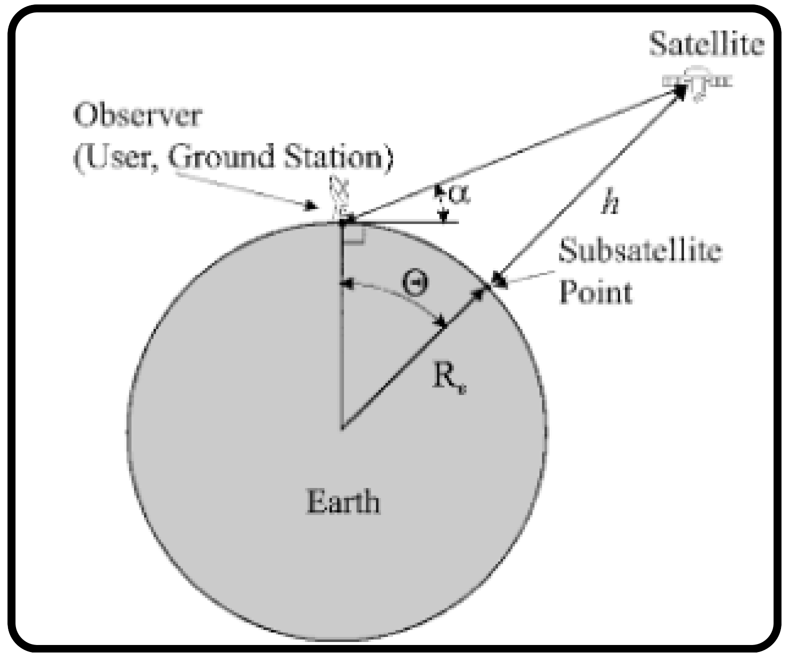


fig. Satellite Geometry Diagram

Latitude & Longitude Estimation

We computed the geographic latitude and longitude of any image pixel using TLE-based satellite positioning and imaging geometry.

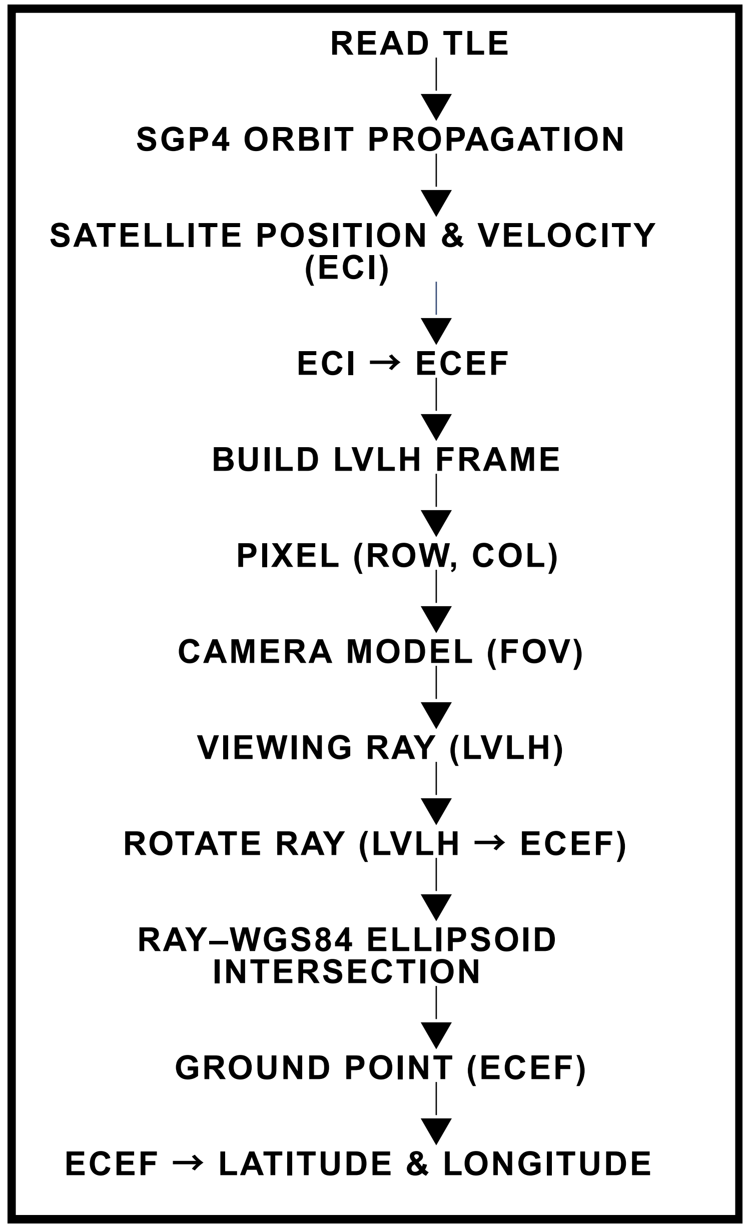
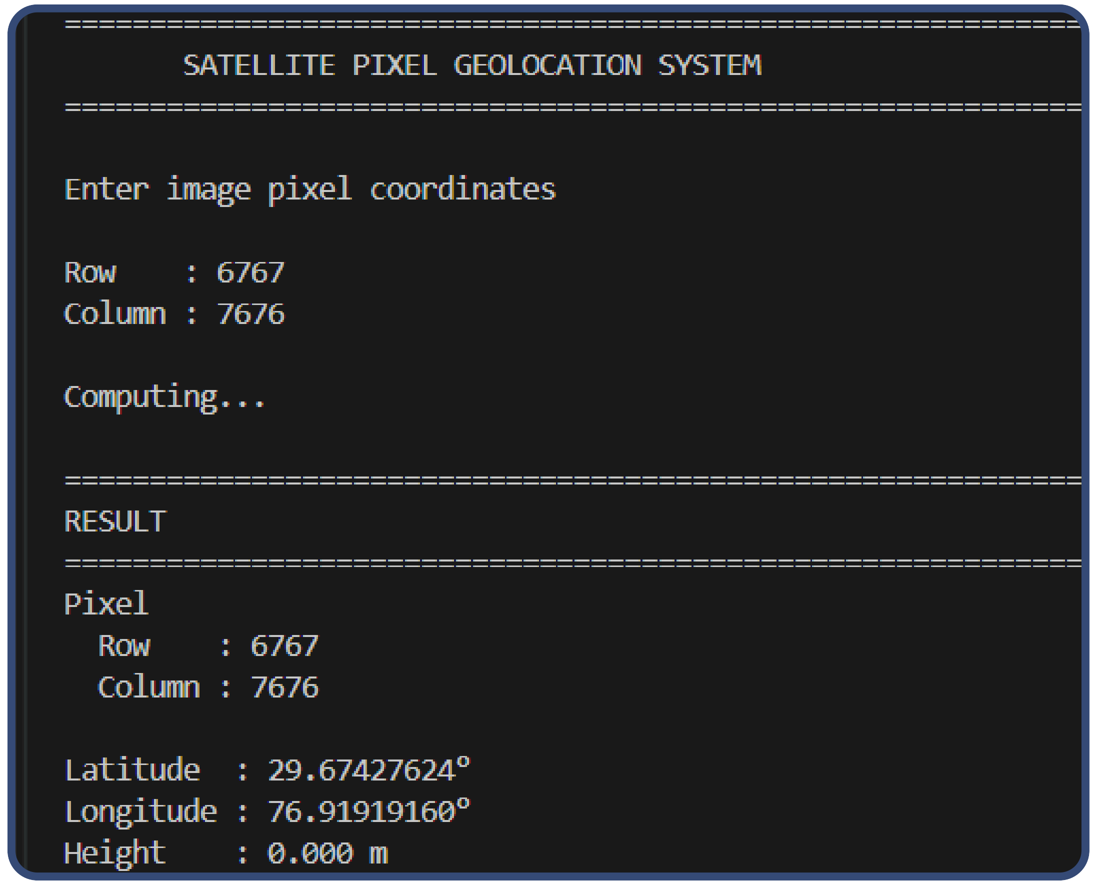
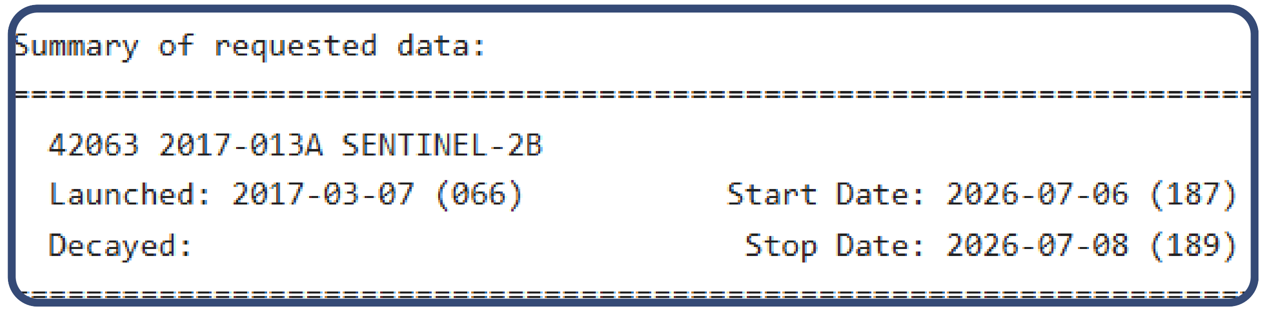


fig. Estimation Pipeline



```
================================================================
     SATELLITE PIXEL GEOLOCATION SYSTEM
================================================================

Enter image pixel coordinates

Row    : 6767
Column : 7676

Computing...

================================================================
RESULT
================================================================
Pixel
  Row    : 6767
  Column : 7676

Latitude  : 29.67427624°
Longitude : 76.91919160°
Height    : 0.000 m
```



```
Summary of requested data:
================================================================
 42063 2017-013A SENTINEL-2B
 Launched: 2017-03-07 (066)          Start Date: 2026-07-06 (187)
 Decayed:                             Stop Date: 2026-07-08 (189)
================================================================
```

fig. Results of Pipeline

Satellite Image Overlay & Coordinate Display

We accurately aligned Sentinel-2 imagery with OpenStreetMap and enable real-time coordinate retrieval.

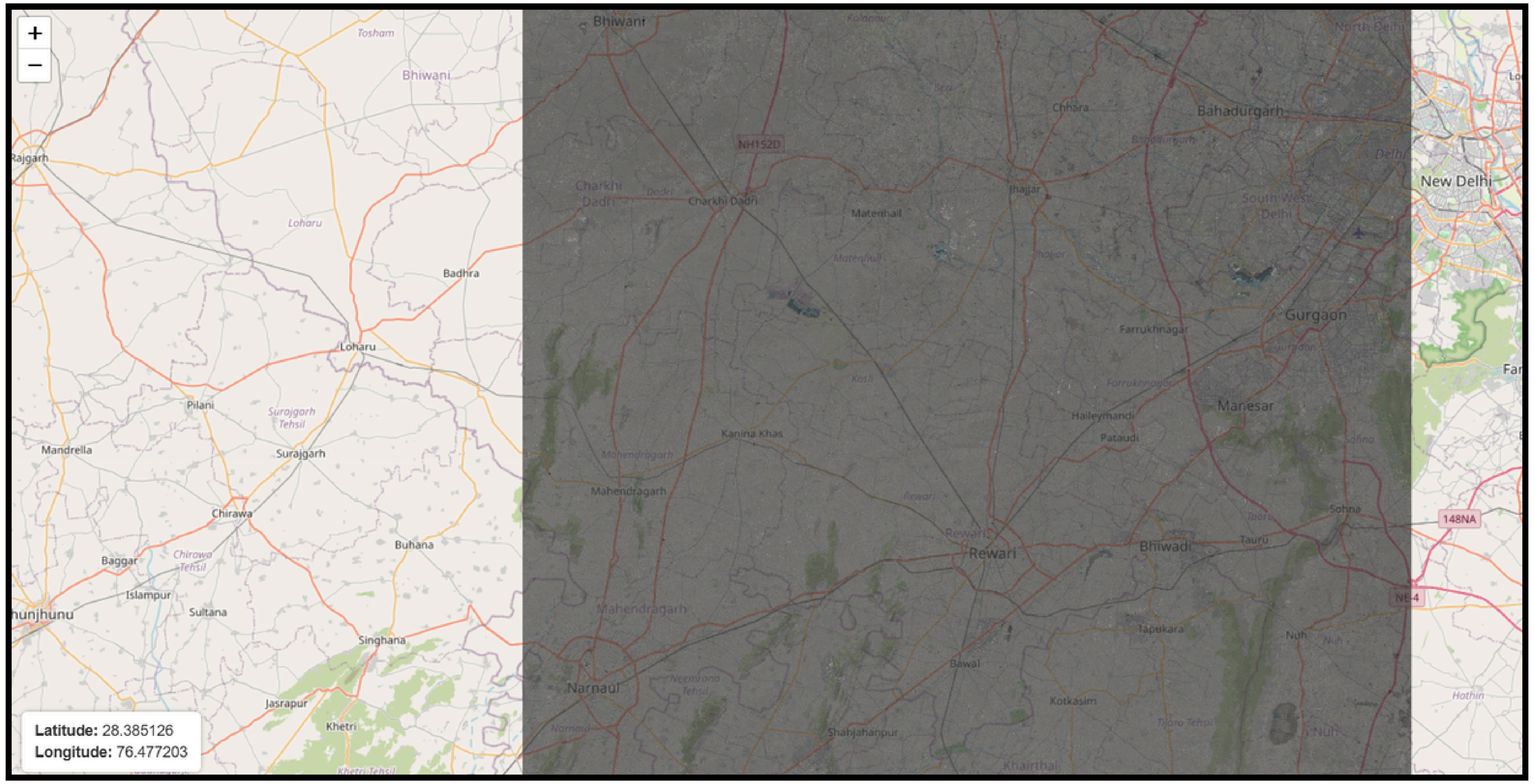


fig. Satellite Imagery overlay aligned directly over a base map (OpenStreetMap)

Sentinel-2 JP2 Image

↓

Read Image & Metadata (Rasterio)

↓

Generate Georeferenced GeoTIFF

↓

Overlay on OpenStreetMap (Folium/Leaflet)

↓

Capture Mouse Position

↓

Display Real-Time Latitude & Longitude

fig. Image Overlay and Coordinate Mapping Pipeline

SUPER RESOLUTION PIPELINE

Our Algorithm

Core Idea:

Instead of treating images as pixels, we treat them as photon measurements. Multiple observations provide additional photon samples that allow recovery of hidden sub-pixel information.

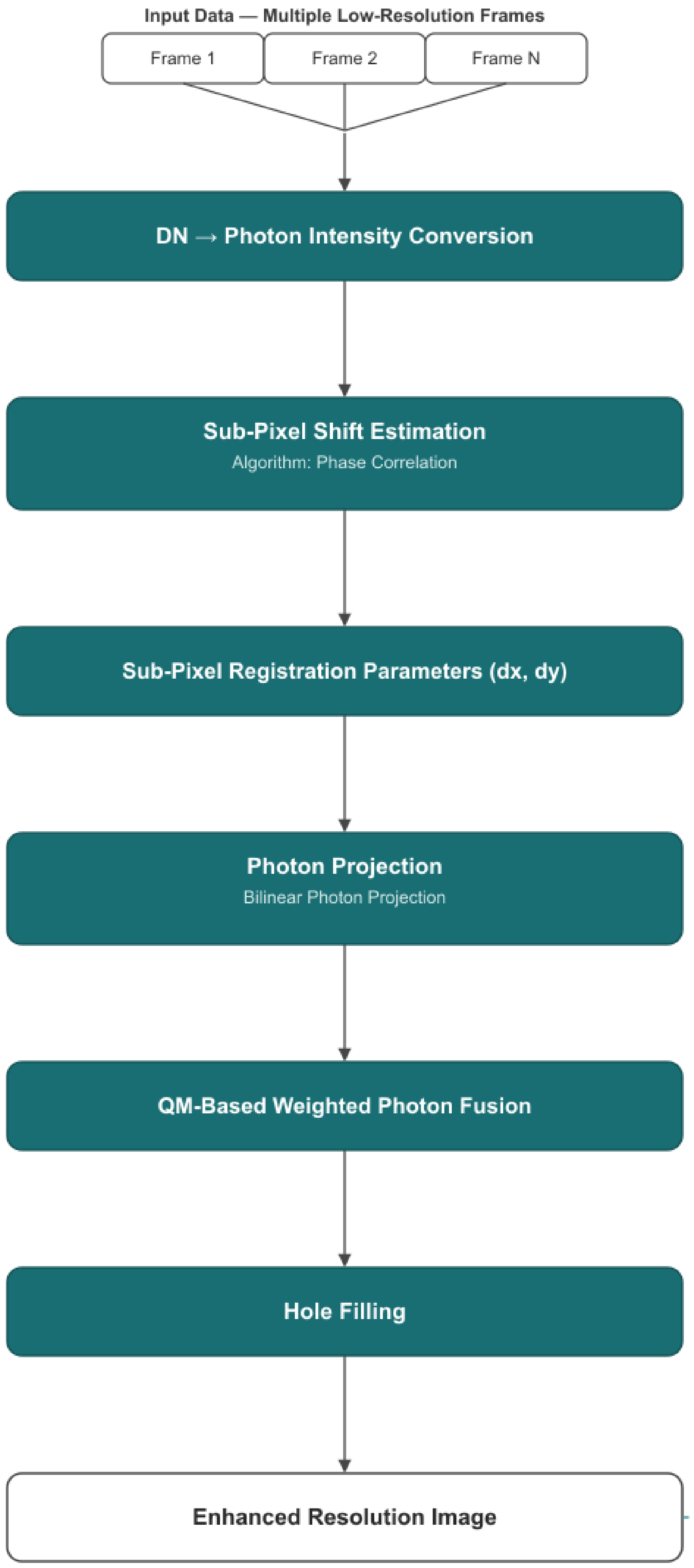
Key Features of our Algorithm:

- Photon-Based Multi-Frame Fusion (Bilinear)
- Sub-Pixel Image Registration
- Quality-Guided Lightweight Reconstruction

Key Formulas Used

$$N_{ph} = \frac{DN - B}{QE \cdot G}$$

$$I_{HR}(u,v) = \frac{\sum_{k=1}^{N} Q_k(u,v)\, P_k(u,v)}{\sum_{k=1}^{N} Q_k(u,v)}$$



DATASET USED

PROBA-V SUPER RESOLUTION IMAGE DATASET

Created by the European Space Agency (ESA) as a real-world benchmark for multi-frame super-resolution algorithms.Features satellite imagery of 74 distinct land regions around the globe, captured in RED and Near-Infrared (NIR) spectral bands.

What the dataset provides us?

- 19 low Low-Resolution (LR) images of same scene (resolution 300m per pixel)
- High-Resolution (HR) image of ground truth (reslution 100m per pixel)

Also Quality mask for both LR and HR images is provided.

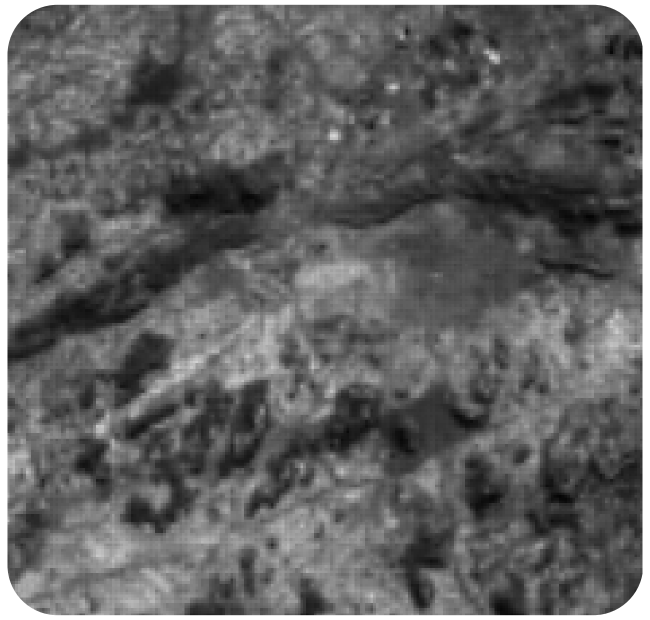


fig. Low Resolution Image (LR001)

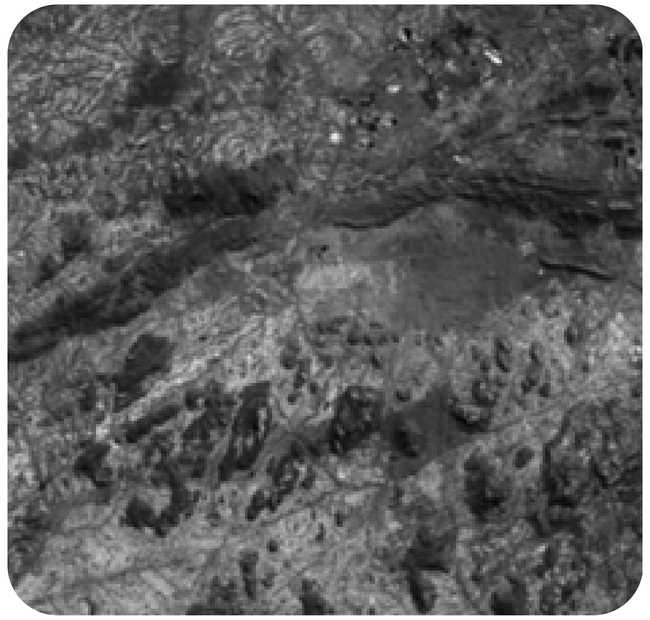


fig. High Resolution Ground Truth (HR)

RESULTS

Results of Our Super Resolution Pipeline

LR000 LR001

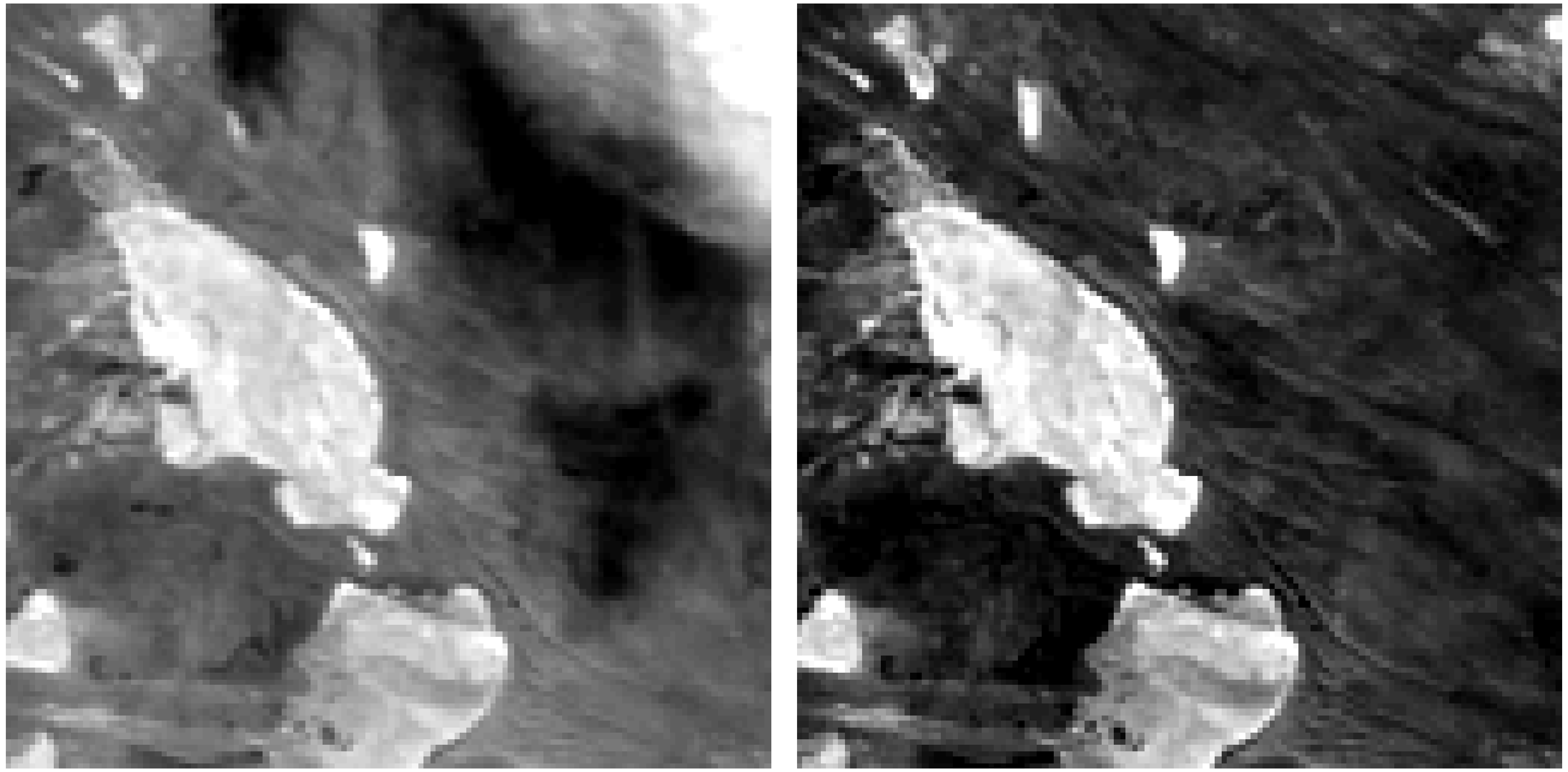


fig. Low Resolution Inputs

Reconstructed Image

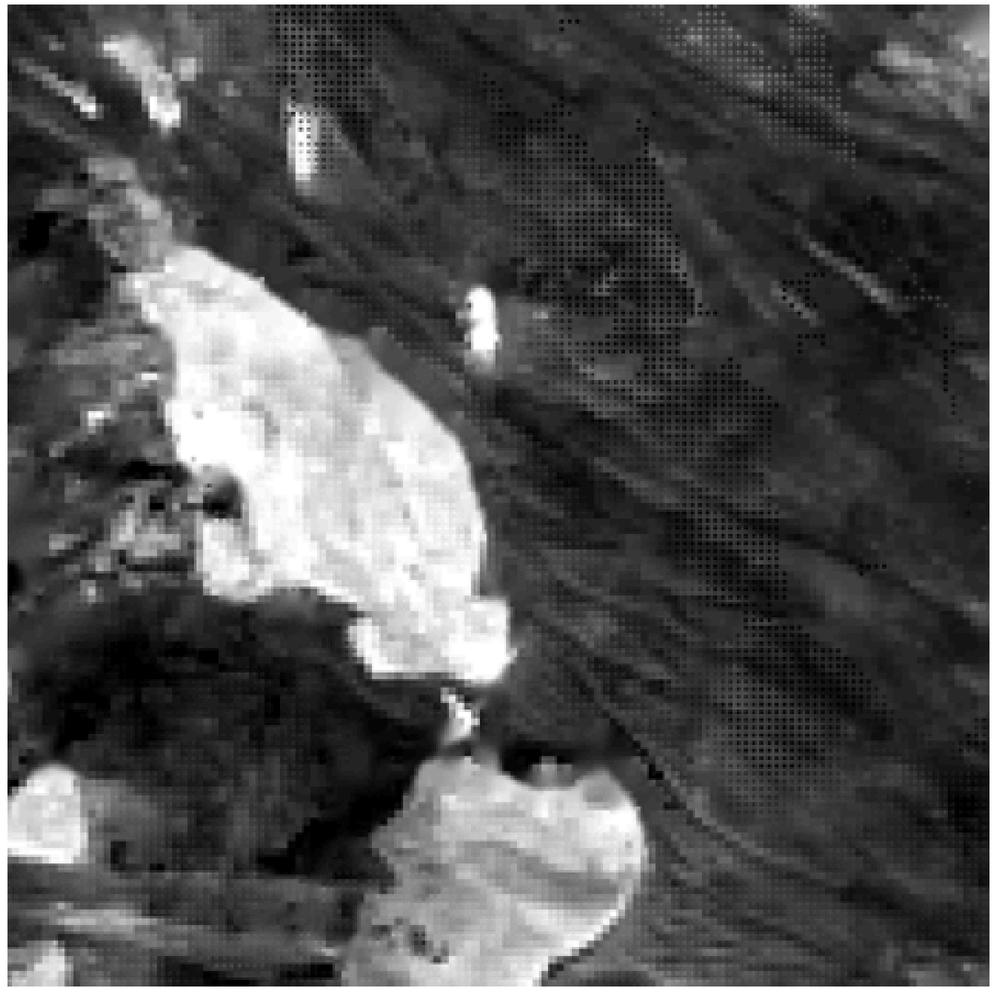


fig. Reconstructed Output

Algorithm Output

DX = -0.0447 PIXELS
DY = 0.0538 PIXELS

PSNR (DB) : 19.48 DB
SSIM : 0.5201

```
Spatial Resolution Metrics
----------------------------------------
Dataset Input GSD (LR)     : 300.0 m/pixel
Dataset Target GSD (HR)    : 100.0 m/pixel
Reconstructed Grid GSD     : 100.0 m/pixel
Effective Spatial Res (SR): 235.00 meters
```

*** The spatial resolution of reconstructed image is improved from 300m/pixel to ~235m/pixel (Edge-Based Gradient Analysis)*

Quantitative Comparison Table

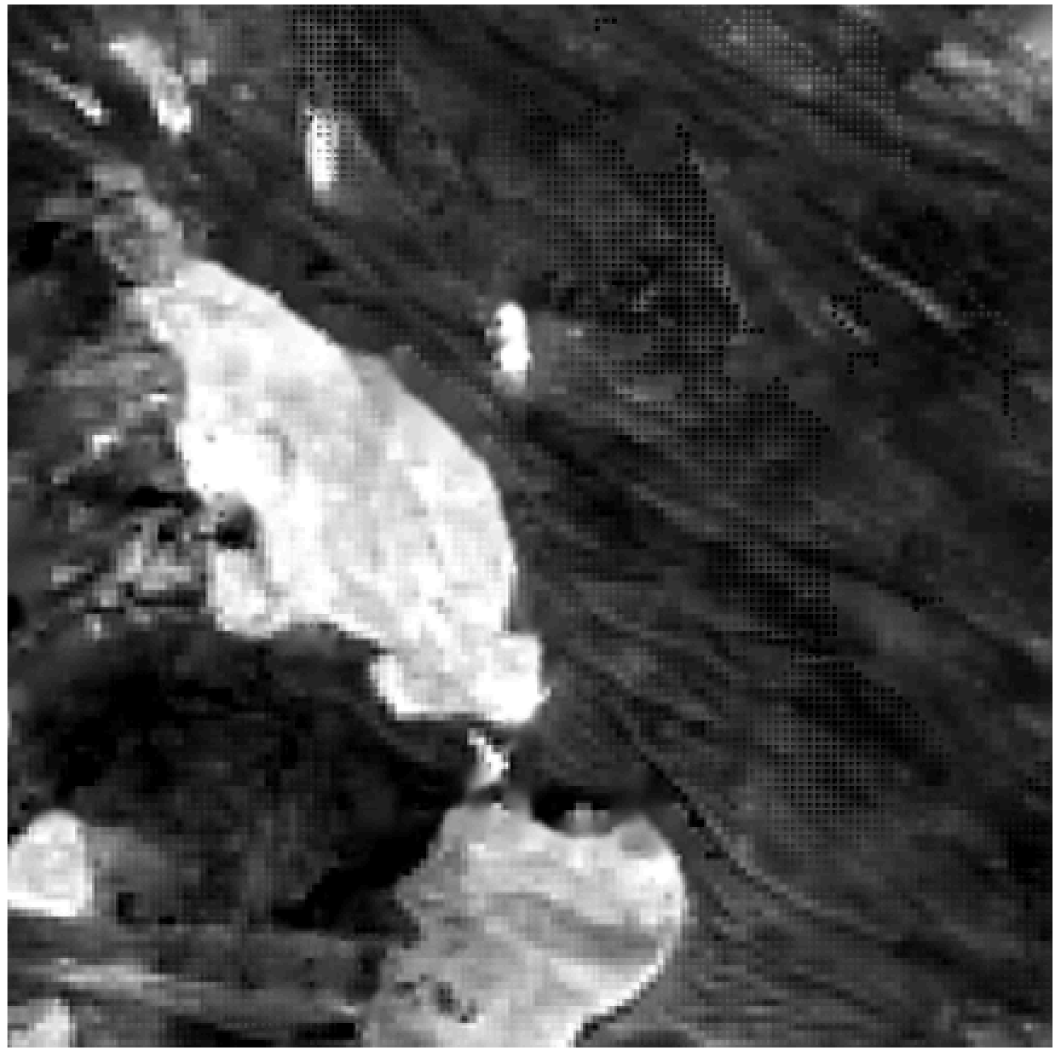


fig. Image Reconstructed by Our Algorithm

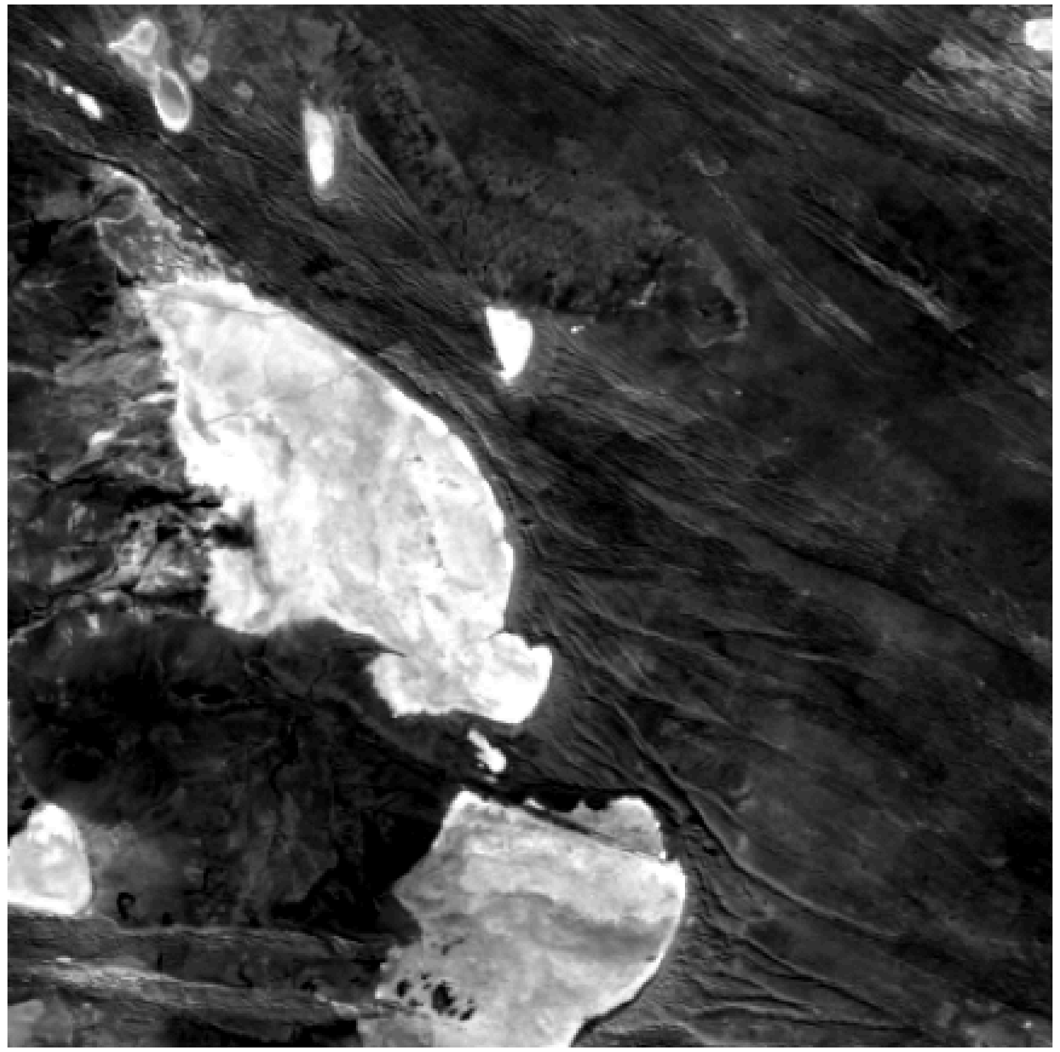


fig. Actual Ground Truth

Method	PSNR (dB)	SSIM
Bicubic Interpolation	17.82	0.412
Shift & Add (Single-frame Average)	18.65	0.468
Proposed Photon-Fusion (Our Algorithm)	19.48	**0.5201**

fig. Comparison to Basic Methods

CONCLUSION

Super Resolution

- Developed a lightweight multi-frame super-resolution pipeline for satellite imaging.
- Combined photon-based fusion and sub-pixel registration to enhance spatial resolution.
- Suitable for efficient onboard implementation in CubeSat OBC systems.

Geospatial Mapping

- Developed an geolocation pipeline to estimate pixel-level latitude and longitude using TLE-based satellite orbit propagation.
- Integrated swath geometry analysis and Ground Sample Distance (GSD) trade-offs for accurate spatial mapping.
- Implemented map alignment and real-time coordinate retrieval by overlaying Sentinel-2 imagery on OpenStreetMap.

Thank You For Listening

All of the documentation and code we created for this project is availabe in this Github repository

GITHUB LINK

Any Questions?